KHOVANOV HOMOLOGY AND EXOTIC SURFACES

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ABSTRACT. IAS/PCMI lecture notes.

CONTENTS

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[1]Based on lectures by Robert Lipshitz [19].

Part 1. **Khovanov Homology and Exotic Surfaces**[2]

1. Khovanov Homology and Link Cobordisms

This section follows Lipshitz's first two lectures [19], Khovanov's construction [12, 13, 14], Lee's deformation [17, 18], and Rasmussen's invariant [27]. For an oriented diagram L, fix

$$N = N_+ + N_-, \qquad A^t := \mathbb{Z}[t][X]/(X^2 - t), \qquad \mathrm{CKh}^t(L)\big|_{t=0} = \mathrm{CKh}(L), \qquad \mathrm{Kh}(L) := H_*(\mathrm{CKh}(L)).$$

For an oriented cobordism $\Sigma\colon L_0 \to L_1$ with chosen movie $\mathscr{M}$, write

$$\Sigma_{\#,\mathscr{M}}\colon \mathrm{CKh}^t(L_0) \longrightarrow q^{\chi(\Sigma)}\mathrm{CKh}^t(L_1), \qquad q^a\mathrm{CKh}^t_{i,j}(L_1) := \mathrm{CKh}^t_{i,j-a}(L_1).$$

Suppress $\mathscr{M}$ only when one movie is fixed.

1.1. **The Khovanov Complex, Gradings, and Invariance.** Cube edges are directed $1 \to 0$, tensor products are over the ground ring, and all gradings use the lasagna convention [24, 28, 22].

Definition 1.1.1 (The deformed Frobenius algebra, [19, Lecture 1, Section 2])**.**

$$A^t := \mathbb{Z}[t][X]/(X^2 - t), \qquad A^t = \mathbb{Z}[t]\cdot 1 \oplus \mathbb{Z}[t]\cdot X, \qquad X^2 = t.$$
$$m\colon A^t \otimes A^t \longrightarrow A^t, \qquad \Delta\colon A^t \longrightarrow A^t \otimes A^t, \qquad 1\colon \mathbb{Z}[t] \longrightarrow A^t, \qquad \varepsilon\colon A^t \longrightarrow \mathbb{Z}[t],$$
$$m(1\otimes 1) = 1, \qquad m(1\otimes X) = m(X\otimes 1) = X, \qquad m(X\otimes X) = t\cdot 1,$$
$$\Delta(1) = 1\otimes X + X\otimes 1, \qquad \Delta(X) = X\otimes X + t(1\otimes 1),$$
$$\varepsilon(1) = 0, \qquad \varepsilon(X) = 1.$$

Definition 1.1.2 (TQFT-graded Frobenius algebra)**.** A *TQFT-graded Frobenius algebra* over a graded commutative ring R is

$$(A, m, \Delta, 1, \varepsilon), \qquad A \in {}_R\mathrm{Mod}_\bullet,$$
$$m\colon A\otimes_R A \to A, \qquad \Delta\colon A \to A\otimes_R A, \qquad 1\colon R \to A, \qquad \varepsilon\colon A \to R,$$
$$\deg(m) = \deg(\Delta) = 1, \qquad \deg(1) = \deg(\varepsilon) = -1,$$
$$m \text{ commutative, associative, and unital}, \qquad \Delta \text{ cocommutative, coassociative, and counital},$$
$$(\mathrm{Id}\otimes m)\circ(\Delta\otimes\mathrm{Id}) = (m\otimes\mathrm{Id})\circ(\mathrm{Id}\otimes\Delta) = \Delta\circ m.$$

Proposition 1.1.3 (The deformed algebra is TQFT-graded Frobenius, [19, Lecture 1, Section 2])**.**

$$\mathrm{gr}_q(1) = -1, \qquad \mathrm{gr}_q(X) = 1, \qquad \mathrm{gr}_q(t) = 4, \qquad (A^t, m, \Delta, 1, \varepsilon) \textit{ is TQFT-graded Frobenius over } \mathbb{Z}[t].$$

Proof. By $\mathbb{Z}[t]$-linearity, the identities reduce to the basis $\{1, X\}$.

$$(\varepsilon\otimes\mathrm{Id})\Delta(1) = 1, \qquad (\varepsilon\otimes\mathrm{Id})\Delta(X) = X, \qquad (\mathrm{Id}\otimes\varepsilon)\Delta = \mathrm{Id}.$$
$$(\Delta\otimes\mathrm{Id})\Delta(1) = 1\otimes X\otimes X + X\otimes 1\otimes X + X\otimes X\otimes 1 + t(1\otimes 1\otimes 1) = (\mathrm{Id}\otimes\Delta)\Delta(1).$$
$$(\Delta\otimes\mathrm{Id})\Delta(X) = X\otimes X\otimes X + t(1\otimes 1\otimes X + 1\otimes X\otimes 1 + X\otimes 1\otimes 1) = (\mathrm{Id}\otimes\Delta)\Delta(X).$$
$$\Delta m(1\otimes 1) = 1\otimes X + X\otimes 1, \qquad \Delta m(1\otimes X) = X\otimes X + t(1\otimes 1),$$
$$\Delta m(X\otimes X) = t(1\otimes X) + t(X\otimes 1) = (\mathrm{Id}\otimes m)(\Delta\otimes\mathrm{Id})(X\otimes X).$$

Cocommutativity supplies the second Frobenius composite, while the quotient-ring multiplication supplies associativity, commutativity, and the unit.

$$\deg(m) = \deg(\Delta) = 1, \qquad \deg(1) = \deg(\varepsilon) = -1,$$
$$\deg(1\otimes 1) = -2 \mapsto \deg(1) = -1, \qquad \deg(X\otimes X) = 2 \mapsto \deg(t\cdot 1) = 4 - 1 = 3,$$
$$\deg(1) = -1 \mapsto \deg\Delta(1) = 0, \qquad \deg(X) = 1 \mapsto \deg\varepsilon(X) = 0.$$
$$\text{the displayed identities and degrees} \overset{1.1.2}{\Longrightarrow} A^t \text{ is TQFT-graded Frobenius.}$$

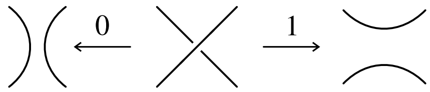
Thus all Frobenius and grading axioms hold. $\square$

[2]Based on lectures by Robert Lipshitz [19].

Fact 1.1.4 (Frobenius–TQFT correspondence, [12, 1])**.** *A TQFT-graded Frobenius algebra A determines a functor*

$$\mathscr{F}_A\colon \mathrm{Cob}(1+1) \longrightarrow {}_R\mathrm{Mod}_\bullet, \qquad \deg\big(\mathscr{F}_A(\Sigma)\big) = -\chi(\Sigma).$$

Definition 1.1.5 (Cube of resolutions, [12, Sections 4.2–4.3], [1], [19, Lecture 1, Section 2])**.** For a crossing c, the 0- and 1-resolutions are fixed by



$$\underline{2}^N := \{0,1\}^N, \qquad v\colon \{c_1,\dots,c_N\} \longrightarrow \{0,1\}, \qquad L_v := \text{complete resolution at } v,$$

$$\mathscr{A}^t(v) := (A^t)^{\otimes|\pi_0(L_v)|}.$$

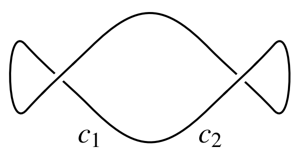
$$v \longrightarrow w \iff \sum_{i=1}^{N}(v_i - w_i) = 1, \qquad v_i \geq w_i,$$

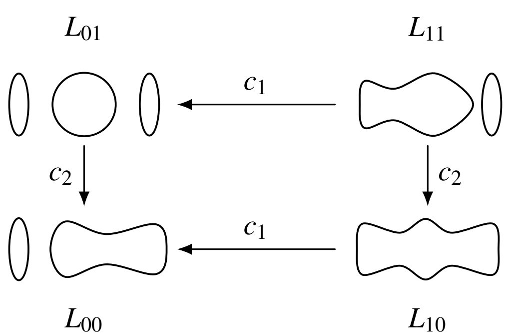
$$\mathscr{A}^t(v \to w) := \begin{cases} m \otimes \mathrm{Id}, & L_v \to L_w \text{ merges two circles,} \\ \Delta \otimes \mathrm{Id}, & L_v \to L_w \text{ splits one circle,} \end{cases}$$

$$\mathscr{A}^t := \big\{\mathscr{A}^t(v), \mathscr{A}^t(v \to w)\big\}_{v,\ v \to w}.$$

Example 1.1.6 (The two-crossing unknot)**.**

$$D := \mathrm{RII}(U), \qquad (c_1, c_2) \text{ ordered as below.}$$





$$\begin{tikzcd}
(A^t)^{\otimes 3} \arrow[d, "\mathrm{Id}\otimes m"'] & (A^t)^{\otimes 2} \arrow[l, "\Delta\otimes\mathrm{Id}"'] \arrow[d, "m"] \\
(A^t)^{\otimes 2} & A^t \arrow[l, "\Delta"']
\end{tikzcd}$$

Proposition 1.1.7 (Commutativity of the Khovanov cube, [12, Section 4.2])**.** *Every two-dimensional face of $\mathscr{A}^t$ commutes.*

Proof. The two routes attach the same pair of disjoint bands in opposite orders.

$$\text{merge–merge} \overset{1.1.2}{\Longrightarrow} m \circ (m \otimes \mathrm{Id}) = m \circ (\mathrm{Id} \otimes m).$$

$$\text{split–split} \overset{1.1.2}{\Longrightarrow} (\Delta \otimes \mathrm{Id}) \circ \Delta = (\mathrm{Id} \otimes \Delta) \circ \Delta.$$

$$\text{merge–split} \overset{1.1.2}{\Longrightarrow} \Delta\circ m = (\mathrm{Id}\otimes m)\circ(\Delta\otimes\mathrm{Id}) = (m\otimes\mathrm{Id})\circ(\mathrm{Id}\otimes\Delta).$$
$$\text{every face commutes} \implies \mathscr{A}^t\colon \underline{2}^N \longrightarrow {}_{\mathbb{Z}[t]}\mathrm{Mod}.$$

Thus the vertex and edge assignment extends to the Khovanov cube functor. $\square$

Definition 1.1.8 (The signed cube)**.**

$$(c_1,\dots,c_N), \qquad v\to w \text{ lowers } v_i = 1 \text{ to } w_i = 0,$$
$$\varepsilon(v,w) := (-1)^{v_1+\cdots+v_{i-1}}, \qquad \mathscr{A}^t_\varepsilon(v\to w) := \varepsilon(v,w)\mathscr{A}^t(v\to w).$$

Lemma 1.1.9 (Anticommutativity of the signed cube)**.** *Every two-dimensional face of $\mathscr{A}^t_\varepsilon$ anticommutes.*

Proof. Fix a face lowering $i<j$ from $v_i = v_j = 1$, and put $s_k := v_1+\cdots+v_{k-1}$.

$$\varepsilon(i,j) = (-1)^{s_i}(-1)^{s_j-1}, \qquad \varepsilon(j,i) = (-1)^{s_j}(-1)^{s_i}.$$
$$\varepsilon(i,j) = -\varepsilon(j,i) \overset{1.1.8}{\Longrightarrow} \text{the signed routes differ by } -1,$$
$$\text{the unsigned routes agree} \overset{1.1.7}{\Longrightarrow} \mathscr{A}^t_\varepsilon(i,j) = -\mathscr{A}^t_\varepsilon(j,i).$$

Thus every signed square anticommutes. $\square$

Definition 1.1.10 (The Khovanov–Lee deformation, [17, 18], [19, Lecture 1, Section 2])**.**

$$\mathrm{CKh}^t(L) := \mathrm{Tot}(\mathscr{A}^t_\varepsilon), \qquad C_j := \bigoplus_{|v|=j}\mathscr{A}^t(v),$$
$$\partial_j(v,x) := \sum_{v\to w}\big(w, \mathscr{A}^t_\varepsilon(v\to w)(x)\big),$$
$$\mathscr{A}^t_\varepsilon(i,j) = -\mathscr{A}^t_\varepsilon(j,i) \overset{1.1.9}{\Longrightarrow} \partial_{j-1}\circ\partial_j = 0.$$
$$\mathrm{CKh}(L) := \mathrm{CKh}^t(L)\otimes_{\mathbb{Z}[t]}\mathbb{Z}[t]/(t), \qquad \mathrm{Kh}(L) := H_*(\mathrm{CKh}(L)).$$

For Example 1.1.6,

$$(A^t)^{\otimes 2} \xleftarrow{\begin{bmatrix}\Delta & \mathrm{Id}\otimes m\end{bmatrix}} A^t\oplus(A^t)^{\otimes 3} \xleftarrow{\begin{bmatrix}-m\\ \Delta\otimes\mathrm{Id}\end{bmatrix}} (A^t)^{\otimes 2}.$$

Proposition 1.1.11 (Independence of the crossing ordering)**.**

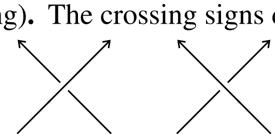
$$(c_1,\dots,c_N)\longmapsto(c_{\sigma(1)},\dots,c_{\sigma(N)}) \implies \mathrm{CKh}^t_\varepsilon(L)\cong\mathrm{CKh}^t_{\varepsilon'}(L) \quad \textit{bigraded.}$$

Proof. Let ε,ε' be the two sign assignments, put $r := \varepsilon/\varepsilon'$, and write $r(e) = (-1)^{\bar r(e)}$.

$$\prod_{e\subset\partial F} r(e) = 1 \overset{1.1.9}{\Longrightarrow} \bar r\in Z^1(\underline{2}^N;\mathbb{Z}/2\mathbb{Z}).$$
$$\underline{2}^N \simeq \{*\} \implies H^1(\underline{2}^N;\mathbb{Z}/2\mathbb{Z}) = 0 \implies \bar r = \delta\nu, \qquad \nu\colon\{0,1\}^N\longrightarrow\mathbb{Z}/2\mathbb{Z}.$$
$$\Phi(v,x) := (-1)^{\nu(v)}(v,x), \qquad \Phi\circ\partial_\varepsilon = \partial_{\varepsilon'}\circ\Phi.$$

Thus Φ is the required bigraded chain isomorphism. $\square$

Definition 1.1.12 (The lasagna bigrading)**.** The crossing signs defining N_+ and N_- are



$+ \qquad -$

$$N = N_+ + N_-, \qquad x\colon \pi_0(L_v) \longrightarrow \{1, X\}, \qquad |v| := \sum_{i=1}^{N} v_i.$$

$$\mathrm{gr}_q(v,x) := \#\{Z \in \pi_0(L_v) \mid x(Z) = X\} - \#\{Z \in \pi_0(L_v) \mid x(Z) = 1\} + |v| + N_+ - 2N_-. \tag{1}$$

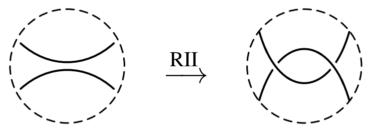
$$\mathrm{gr}_h(v,x) := |v| - N_-. \tag{2}$$

$$\mathrm{gr}_q\big(t^k(v,x)\big) = \mathrm{gr}_q(v,x) + 4k, \qquad \mathrm{gr}_h\big(t^k(v,x)\big) = \mathrm{gr}_h(v,x),$$

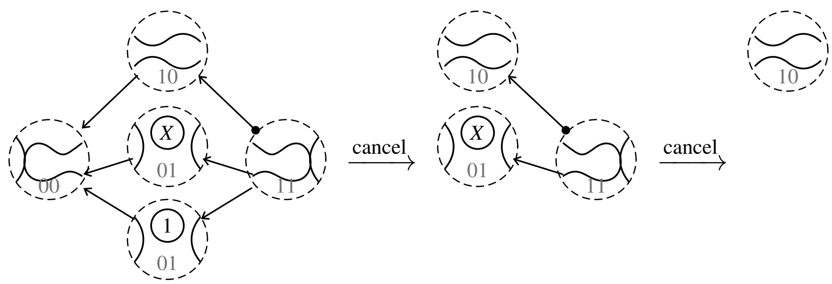
$$\deg(m) = \deg(\Delta) = 1 \overset{1.1.3}{\Longrightarrow} \mathrm{gr}_q(\partial(v,x)) = \mathrm{gr}_q(v,x), \qquad \mathrm{gr}_h(\partial(v,x)) = \mathrm{gr}_h(v,x) - 1.$$

Proposition 1.1.13 (Second Reidemeister invariance, [12, Section 5], [1], [19, Lecture 1, Section 3])**.**

$$D' = \mathrm{RII}(D) \implies \mathrm{CKh}^t(D') \simeq \mathrm{CKh}^t(D) \quad \textit{bigraded}.$$



Proof. The RII cube has two acyclic summands, cancelled successively below; a tail dot records a minus sign, and the 01-resolution occurs once for each label on the small circle.



$$|v| \mapsto |v| + 1, \qquad N_+ \mapsto N_+ + 1, \qquad N_- \mapsto N_- + 1.$$

$$\big(|v|, N_+, N_-\big) \overset{1.1.12}{\Longrightarrow} \mathrm{gr}_q \mapsto \mathrm{gr}_q + 1 + 1 - 2 = \mathrm{gr}_q,$$

$$\big(|v|, N_-\big) \overset{1.1.12}{\Longrightarrow} \mathrm{gr}_h \mapsto \mathrm{gr}_h + 1 - 1 = \mathrm{gr}_h.$$

Thus both cancellations preserve the bigrading and leave $\mathrm{CKh}^t(D)$. $\square$

Proposition 1.1.14 (First Reidemeister invariance, [12, Section 5], [1], [19, Lecture 1, Section 3])**.**

$$D' = \mathrm{RI}(D) \implies \mathrm{CKh}^t(D') \simeq \mathrm{CKh}^t(D) \quad \textit{bigraded}.$$

Proof. For a positive kink, the new edge is the split map on the circle through the kink.

$$\mathrm{CKh}^t(D) \xrightarrow{\Delta_Z} \mathrm{CKh}^t(D) \otimes_{\mathbb{Z}[t]} A^t.$$

$$\Delta(1) = 1 \otimes X + X \otimes 1, \qquad \Delta(X) = X \otimes X + t(1 \otimes 1).$$

$$X \otimes 1 \equiv -1 \otimes X, \qquad X \otimes X \equiv -t(1 \otimes 1) \overset{1.1.1}{\Longrightarrow} \mathrm{coker}(\Delta) = \langle 1 \otimes 1, 1 \otimes X \rangle_{\mathbb{Z}[t]}.$$

$$\#1 \mapsto \#1 + 1, \qquad N_+ \mapsto N_+ + 1 \overset{1.1.12}{\Longrightarrow} \mathrm{gr}_q \mapsto \mathrm{gr}_q - 1 + 1 = \mathrm{gr}_q,$$

$$(|v|, N_-) \mapsto (|v|, N_-) \overset{1.1.12}{\Longrightarrow} \mathrm{gr}_h \mapsto \mathrm{gr}_h.$$

Cancelling the split injection leaves its free cokernel, identified with $\mathrm{CKh}^t(D)$ by the surviving small-circle label 1.

$$m\big((X \cdot y) \otimes 1 - y \otimes X\big) = 0, \qquad \ker(m) \cong A^t, \qquad y \longmapsto (X \cdot y) \otimes 1 - y \otimes X.$$

$$\mathrm{gr}_{\mathrm{int}}(y) \mapsto \mathrm{gr}_{\mathrm{int}}(y) + 1, \qquad |v| \mapsto |v| + 1, \qquad N_- \mapsto N_- + 1.$$

$$\left(\mathrm{gr}_{\mathrm{int}}, |v|, N_-\right) \overset{1.1.12}{\Longrightarrow} \mathrm{gr}_q \mapsto \mathrm{gr}_q + 1 + 1 - 2 = \mathrm{gr}_q,$$
$$\left(|v|, N_-\right) \overset{1.1.12}{\Longrightarrow} \mathrm{gr}_h \mapsto \mathrm{gr}_h + 1 - 1 = \mathrm{gr}_h.$$

For a negative kink, cancelling the unit summand of the merge leaves this kernel, so the same bigraded equivalence follows. □

Theorem 1.1.15 (Reidemeister invariance, [12, Theorem 1 and Section 5], [1], [19, Lecture 1, Section 3])**.**

$$D \simeq D' \implies \mathrm{CKh}^t(D) \simeq \mathrm{CKh}^t(D') \implies \mathrm{Kh}(D) \cong \mathrm{Kh}(D') \quad \textit{bigraded}.$$

Convention	Type	$\mathrm{gr}(\cdot X)$	gr_q	gr_h	Used in
Khovanov original	Cohomological $(0 \to 1)$	-2	$\#1 - \#X + \lvert v\rvert + N_+ - 2N_-$	$\lvert v\rvert - N_-$	[12, 1, 16, 30, 10]
Khovanov tangles	Cohomological $(0 \to 1)$	2	$\#X - \#1 - \lvert v\rvert - N_+ + 2N_-$	$\lvert v\rvert - N_-$	[13]
Lasagna	Homological $(1 \to 0)$	2	$\#X - \#1 + \lvert v\rvert + N_+ - 2N_-$	$\lvert v\rvert - N_-$	[24, 28, 22]

TABLE 1. The principal grading conventions used for Khovanov complexes.

Fact 1.1.16 (Mirror and lasagna gradings, [12, Section 7.3, Propositions 31–32, and Corollary 11], [27, Proposition 3.9], [24, 28, 22])**.**

$$\mathrm{Kh}^{i,j}_{\mathrm{las}}(L) \cong \mathrm{Kh}^{-i,-j}_{\mathrm{orig}}(mL), \qquad -N_- \le \mathrm{gr}_h \le N_+.$$

Comparing the lasagna and original conventions for the same link over $\mathbb{Z}$, duality and the universal coefficient theorem preserve free bidegrees and lower torsion by one homological degree.

$$s_{\mathrm{las}}(K) = s_{\mathrm{Ras}}(K), \qquad s(mK) = -s(K).$$

Definition 1.1.17 (Lee specialization, [17, 18, 27])**.**

$$A^t/(t-1) = \mathbb{Z}[X]/(X^2-1), \qquad \mathrm{CKh}_{\mathrm{Lee}}(L;\mathbb{Q}) := \mathrm{CKh}^t(L) \otimes_{\mathbb{Z}[t]} \mathbb{Q}[t]/(t-1).$$
$$t = 1 \overset{1.1.1}{\Longleftrightarrow} X^2 = 1.$$

The integral algebra is filtered, and the rational complex is its $t = 1$ specialization.

1.2. **Movies, Functoriality, and Rasmussen's Invariant.** A chosen movie $\mathscr{M}$ of a smooth oriented cobordism carries the map

$$\Sigma_{\#,\mathscr{M}} \colon \mathrm{CKh}^t_{i,j}(L_0) \longrightarrow \mathrm{CKh}^t_{i,j-\chi(\Sigma)}(L_1).$$

The sign and the passage from $[0,1] \times \mathbb{R}^3$ to $[0,1] \times S^3$ are treated in Section 2.

Definition 1.2.1 (Link cobordisms, [12, 6], [19, Lecture 2, Section 2.1])**.**

$$\Sigma \colon L_0 \longrightarrow L_1, \qquad \Sigma \subset [0,1] \times \mathbb{R}^3, \qquad \partial\Sigma = (-\{0\} \times L_0) \sqcup (\{1\} \times L_1).$$

For some $\delta > 0$, require

$$\Sigma \cap ([0,\delta] \times \mathbb{R}^3) = [0,\delta] \times L_0, \qquad \Sigma \cap ([1-\delta,1] \times \mathbb{R}^3) = [1-\delta,1] \times L_1.$$
$$\Sigma_1 \colon L_0 \to L_1, \qquad \Sigma_2 \colon L_1 \to L_2 \implies \Sigma_2 \circ \Sigma_1 \colon L_0 \to L_2,$$
$$(L_s)_{s\in[0,1]} \longmapsto \bigcup_{s\in[0,1]} \{s\} \times L_s.$$

Composition uses concatenation and rescaling, isotopies fix the displayed boundary, and the standard orientation-reversing identification sends the incoming boundary to $-mL_0$.

Definition 1.2.2 (Movies and elementary cobordisms, [6], [11, Section 3.1], [19, Lecture 2, Section 2.1])**.**

$$\mathscr{E} := \{\text{planar isotopy}, \text{Reidemeister move}, \text{birth}, \text{death}, \text{saddle}\}.$$

A cobordism in $\mathscr{E}$ is *elementary*. A *movie* is

$$D_0 \xrightarrow{\Sigma_1} D_1 \xrightarrow{\Sigma_2} \cdots \xrightarrow{\Sigma_n} D_n, \qquad \Sigma_i \in \mathscr{E}.$$

The diagrams D_i are its frames.

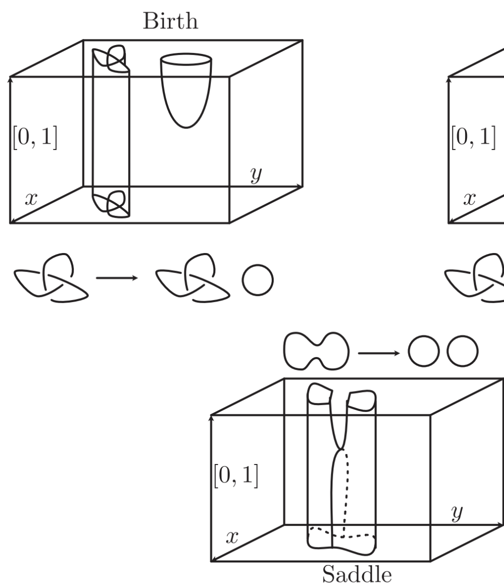


FIGURE 1. The elementary birth, death, and saddle cobordisms and their effects on link diagrams[19].

Example 1.2.3 (Explicit elementary surfaces, [19, Lecture 2, Section 2.1])**.** Fix a standard ball B, coordinates (t,x,y,z), and the diagram plane $z=0$. Away from B, use the constant trace; inside B, set

$$\Sigma_{\mathrm{b}} \cap B = \{t = \tfrac{1}{2} + x^2 + y^2,\ z = 0\} \cap B, \qquad \Sigma_{\mathrm{d}} \cap B = \{t = \tfrac{1}{2} - x^2 - y^2,\ z = 0,\ t \geq 0\} \cap B,$$

$$\Sigma_{\mathrm{s}} \cap B = \{x^2 - y^2 = t - \tfrac{1}{2},\ z = 0,\ x^2 + y^2 \leq 1\} \cap B.$$

Proof. The height restrictions give the three Morse models directly.

$$\operatorname{rank}\begin{bmatrix} 1 & \mp 2x & \mp 2y & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = 2, \qquad \operatorname{rank}\begin{bmatrix} -1 & 2x & -2y & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = 2,$$

$$t|_{\Sigma_{\mathrm{b}}} = \tfrac{1}{2} + x^2 + y^2 \implies \operatorname{ind}_0(t|_{\Sigma_{\mathrm{b}}}) = 0, \qquad t|_{\Sigma_{\mathrm{d}}} = \tfrac{1}{2} - x^2 - y^2 \implies \operatorname{ind}_0(t|_{\Sigma_{\mathrm{d}}}) = 2,$$

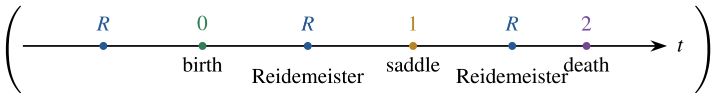
$$t|_{\Sigma_{\mathrm{s}}} = \tfrac{1}{2} + x^2 - y^2 \implies \operatorname{ind}_0(t|_{\Sigma_{\mathrm{s}}}) = 1.$$

Thus these are the birth, death, and saddle models of Figure 1. □

Lemma 1.2.4 (Elementary decomposition, [6], [11, Section 3.1], [19, Lecture 2, Section 2.1])**.**

$$\Sigma\colon L_0 \longrightarrow L_1 \implies \Sigma \simeq_\partial \Sigma_n \circ \cdots \circ \Sigma_1, \qquad \Sigma_i \in \mathscr{E}.$$

Proof. Perturb the height and planar projection to generic position, and apply the Morse lemma at each critical point.



$$\Longrightarrow (0,1,2) \longleftrightarrow (\mathrm{b},\mathrm{s},\mathrm{d}),$$

$$t|_\Sigma \text{ Morse with distinct critical values} \overset{1.2.2}{\Longrightarrow} \Sigma \simeq \Sigma_n \circ \cdots \circ \Sigma_1,$$
$$\Sigma_i \in \{\text{planar isotopy}, \text{Reidemeister move}, \text{birth}, \text{death}, \text{saddle}\}.$$

Thus the generic movie is an elementary decomposition of Σ. □

Remark 1.2.5 (Topological movies). Smoothness supplies the generic Morse movie used in Lemma 1.2.4; no such argument applies directly to topologically locally flat cobordisms.

Lemma 1.2.6 (Births are tensoring)**.**

$$U \cap D = \varnothing, \qquad N(U) = 0 \implies \mathrm{CKh}^t(D \sqcup U) \cong \mathrm{CKh}^t(D) \otimes_{\mathbb{Z}[t]} A^t.$$

Proof. The extra component supplies one unchanged tensor factor at every cube vertex.

$$(D \sqcup U)_v = D_v \sqcup U, \qquad \mathscr{A}^t_{D \sqcup U}(v) = \mathscr{A}^t_D(v) \otimes_{\mathbb{Z}[t]} A^t,$$
$$\mathscr{A}^t_{D \sqcup U}(v \to w) = \mathscr{A}^t_D(v \to w) \otimes \mathrm{Id}_{A^t}, \qquad \varepsilon_{D \sqcup U}(v,w) = \varepsilon_D(v,w),$$
$$\left(\mathscr{A}^t_{D \sqcup U}, \varepsilon_{D \sqcup U}\right) \overset{1.1.5, 1.1.10}{\Longrightarrow} \mathrm{Tot}(\mathscr{A}^t_{D \sqcup U}) = \mathrm{Tot}(\mathscr{A}^t_D) \otimes_{\mathbb{Z}[t]} A^t.$$

Thus totalization gives the asserted bigraded isomorphism. □

Theorem 1.2.7 (Weak functoriality, [12, Section 6.3], [19, Lecture 2, Section 2.2])**.**

$$\Sigma \colon L_0 \longrightarrow L_1 \overset{1.2.4}{\Longrightarrow} \exists \mathscr{M} = (\Sigma_1, \ldots, \Sigma_n) \textit{ an elementary movie for } \Sigma.$$
$$\mathscr{M} \textit{ an elementary movie for } \Sigma \implies \Sigma_{\#,\mathscr{M}} \colon \mathrm{CKh}^t_{i,j}(L_0) \longrightarrow \mathrm{CKh}^t_{i,j-\chi(\Sigma)}(L_1).$$

Proof. By Lemma 1.2.4, choose an elementary movie and compose its local maps.

$$(\text{planar isotopy or Reidemeister move})_\# \overset{1.1.15}{\Longrightarrow} \text{bigraded homotopy equivalence},$$
$$(\text{birth})_\#(x) = x \otimes 1 \overset{1.2.6}{\Longrightarrow} x \otimes 1 \in \mathrm{CKh}^t(D \sqcup U),$$
$$(\text{death})_\#(x \otimes y) = \varepsilon(y)x, \qquad S_v := (\text{saddle})_{\#,v} = m \text{ or } \Delta,$$
$$\mathscr{A}^t_{D'}(v \to w) \circ S_v = S_w \circ \mathscr{A}^t_D(v \to w) \overset{1.1.3}{\Longrightarrow} \partial_{D'} S = S \partial_D,$$
$$\Sigma = \Sigma_n \circ \cdots \circ \Sigma_1 \overset{1.2.2}{\Longrightarrow} \Sigma_{\#,\mathscr{M}} = (\Sigma_n)_\# \circ \cdots \circ (\Sigma_1)_\#,$$
$$\deg(\Sigma_i)_\# = (0, -\chi(\Sigma_i)) \overset{1.1.4}{\Longrightarrow} \deg \Sigma_{\#,\mathscr{M}} = (0, -\chi(\Sigma)).$$

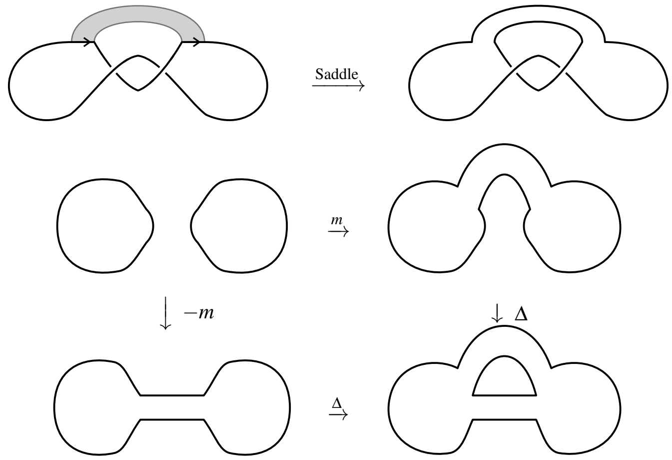
Thus the elementary TQFT maps define the asserted movie map. □

Theorem 1.2.8 (Connected Lee specialization, [27, Corollary 4.2])**.**

$$\mathbb{Q}_{t=1} := \mathbb{Q}[t]/(t-1), \qquad \mathrm{CKh}_{\mathrm{Lee}}(K) := \mathrm{CKh}^t(K) \otimes_{\mathbb{Z}[t]} \mathbb{Q}_{t=1}.$$

$$\Sigma \colon K_0 \longrightarrow K_1, \qquad |\pi_0(\Sigma)| = 1 \implies H(\Sigma_\#) \colon H\big(\mathrm{CKh}_{\mathrm{Lee}}(K_0)\big) \xrightarrow{\cong} H\big(\mathrm{CKh}_{\mathrm{Lee}}(K_1)\big). \tag{3}$$

Example 1.2.9 (A nonorientable saddle)**.** The saddle below joins equally oriented parallel strands and induces the displayed cube map. It is not a morphism of Definition 1.2.1.



Corollary 1.2.10 (Free summands over the polynomial ring, [18, Theorem 4.2 and Proposition 4.3], [27, 19])**.**

$$\mathrm{Kh}^t(K;\mathbb{Q}[t]) \cong \mathbb{Q}[t]_{(0,p)} \oplus \mathbb{Q}[t]_{(0,q)} \oplus T, \qquad T := \mathrm{tors}_{\mathbb{Q}[t]}\big(\mathrm{Kh}^t(K;\mathbb{Q}[t])\big), \qquad (\cdot,\cdot) = (\mathrm{gr}_h, \mathrm{gr}_q).$$

Proof. Choose a connected cobordism $K \to U$, use the graded structure theorem, and specialize at $t = 1$.

$$C_J := \mathrm{CKh}^t(J) \otimes_{\mathbb{Z}[t]} \mathbb{Q}[t] \quad (J = K, U), \qquad H(C_K) = \mathrm{Kh}^t(K;\mathbb{Q}[t]),$$

$$H(C_K) \cong \bigoplus_{a=1}^{r} \mathbb{Q}[t]_{(i_a, j_a)} \oplus \bigoplus_{b=1}^{s} \big(\mathbb{Q}[t]/(t^{n_b})\big)_{(u_b, k_b)},$$

$$0 \longrightarrow \mathrm{Kh}_i^t(K;\mathbb{Q}[t]) \otimes_{\mathbb{Q}[t]} \mathbb{Q}_{t=1} \longrightarrow H_i\big(C_K \otimes_{\mathbb{Q}[t]} \mathbb{Q}_{t=1}\big) \longrightarrow \mathrm{Tor}_1^{\mathbb{Q}[t]}\big(\mathrm{Kh}_{i-1}^t(K;\mathbb{Q}[t]), \mathbb{Q}_{t=1}\big) \longrightarrow 0,$$

$$T \otimes_{\mathbb{Q}[t]} \mathbb{Q}_{t=1} = 0, \qquad \mathrm{Tor}_1^{\mathbb{Q}[t]}(T, \mathbb{Q}_{t=1}) = 0,$$

$$H\big(C_K \otimes_{\mathbb{Q}[t]} \mathbb{Q}_{t=1}\big) \overset{1.2.8}{\Longrightarrow} H\big(C_U \otimes_{\mathbb{Q}[t]} \mathbb{Q}_{t=1}\big) \cong \mathbb{Q}^2,$$

$$\mathrm{rank}_{\mathbb{Q}[t]} \mathrm{Kh}^t(K;\mathbb{Q}[t]) = 2, \qquad H_i\big(C_K \otimes_{\mathbb{Q}[t]} \mathbb{Q}_{t=1}\big) = 0 \quad (i \neq 0),$$

$$r = 2, \qquad i_1 = i_2 = 0, \qquad (j_1, j_2) = (p, q).$$

Thus precisely two free summands survive, both in homological grading zero. □

Proposition 1.2.11 (Rasmussen's gap, [27, Proposition 3.3], [19])**.** *Under the t-deformation, p, q are the two Lee filtration degrees.*

$$M_K := \mathrm{Kh}^t(K;\mathbb{Q}[t])/\mathrm{tors}\big(\mathrm{Kh}^t(K;\mathbb{Q}[t])\big), \qquad M_K \cong \mathbb{Q}[t]_{(0,p)} \oplus \mathbb{Q}[t]_{(0,q)}, \qquad |p - q| = 2.$$

Definition 1.2.12 (Rasmussen's s-invariant)**.**

$$M_K \cong \mathbb{Q}[t]_{(0,p)} \oplus \mathbb{Q}[t]_{(0,q)} \overset{1.2.11}{\Longrightarrow} |p - q| = 2, \qquad s(K) := \frac{p+q}{2},$$

$$|p - q| = 2 \iff \{p, q\} = \{s(K) - 1, s(K) + 1\}.$$

Corollary 1.2.13 (Cobordism genus bound, [27, Corollary 4.2 and proof of Theorem 1])**.**

$$\Sigma\colon K_1 \longrightarrow K_2, \qquad |\pi_0(\Sigma)| = 1, \qquad g(\Sigma) = g \implies |s(K_1) - s(K_2)| \leq 2g.$$

Proof. By Corollary 1.2.10 and Proposition 1.2.11, pass to the free quotients and write the cobordism map as a homogeneous matrix over $\mathbb{Q}[t]$.

$$M_i := \mathrm{Kh}^t(K_i;\mathbb{Q}[t])/\mathrm{tors}\big(\mathrm{Kh}^t(K_i;\mathbb{Q}[t])\big),$$
$$M_i \overset{1.2.10,1.2.11,1.2.12}{\Longrightarrow} M_i \cong \mathbb{Q}[t]_{(0,s(K_i)-1)} \oplus \mathbb{Q}[t]_{(0,s(K_i)+1)},$$
$$\chi(\Sigma) = -2g, \qquad \deg_q(\Sigma_\#|_{M_1}) = 2g,$$
$$|\pi_0(\Sigma)| = 1 \overset{1.2.8}{\Longrightarrow} (\Sigma_\#|_{M_1})_{t=1} \in \mathrm{GL}_2(\mathbb{Q}),$$
$$\exists \varepsilon \in \{-1,1\},\ n \geq 0,\ c \in \mathbb{Q}^\times, \qquad M_{+,\varepsilon} = c \cdot t^n, \qquad s(K_2)+1+4n = s(K_1)+\varepsilon+2g,$$
$$s(K_2) \leq s(K_1)+2g, \qquad \overline{\Sigma}\colon K_2 \to K_1 \implies s(K_1) \leq s(K_2)+2g.$$

Thus the two homogeneous-degree inequalities give the claimed bound. □

Proposition 1.2.14 (The s-invariant of positive torus knots, [27, Theorem 4 and Section 5.2])**.**

$$\gcd(p,q) = 1, \qquad p,q > 0 \implies s(T_{p,q}) = (p-1)(q-1).$$

Proof. Apply Rasmussen's positive-diagram formula to the standard braid closure.

$$T_{p,q} = \widehat{(\sigma_1 \cdots \sigma_{p-1})^q}, \qquad N = q(p-1), \qquad \#\{\text{Seifert circles}\} = p,$$
$$D \text{ positive} \implies s(T_{p,q}) = N - \#\{\text{Seifert circles}\} + 1,$$
$$N - p + 1 = q(p-1) - p + 1 = (p-1)(q-1).$$

Thus the positive-diagram formula gives the stated invariant. □

Corollary 1.2.15 (Milnor conjecture, [23, Theorems 6.5, 7.2, and 9.1], [15], [27, Corollary 1])**.**

$$\gcd(p,q) = 1, \qquad p,q > 0 \implies g_4(T_{p,q}) = \frac{(p-1)(q-1)}{2}.$$

Proof. Puncture any bounding surface for the lower bound and use the Milnor fiber for the upper bound.

$$F \subset B^4, \qquad \partial F = T_{p,q}, \qquad g(F) = g \implies F \setminus D^2\colon T_{p,q} \to U,$$
$$F \setminus D^2\colon T_{p,q} \to U \overset{1.2.13}{\Longrightarrow} 2g \geq |s(T_{p,q}) - s(U)|, \qquad s(U) = 0, \overset{1.2.14}{\Longrightarrow} g \geq \frac{(p-1)(q-1)}{2},$$
$$\mu(x^p + y^q) = \dim_{\mathbb{C}} \frac{\mathbb{C}[x,y]}{(x^{p-1}, y^{q-1})} = (p-1)(q-1), \qquad \chi(F_{\mathrm{Milnor}}) = 1 - \mu = p - q(p-1),$$
$$g(F_{\mathrm{Milnor}}) = \frac{1-\chi(F_{\mathrm{Milnor}})}{2} = \frac{(p-1)(q-1)}{2} \implies g_4(T_{p,q}) \leq \frac{(p-1)(q-1)}{2}.$$

Thus the Milnor fiber realizes the lower bound. □

Theorem 1.2.16 (Carter–Saito movie moves, [6], [11, Section 4.2])**.**

$$\Sigma \simeq_\partial \Sigma' \iff \mathscr{M}(\Sigma) \sim_{\mathrm{CS}} \mathscr{M}(\Sigma'),$$
$$\sim_{\mathrm{CS}} := \{\textit{fifteen local movie moves}, \textit{planar movie isotopy}, \textit{distant commutation}\}.$$

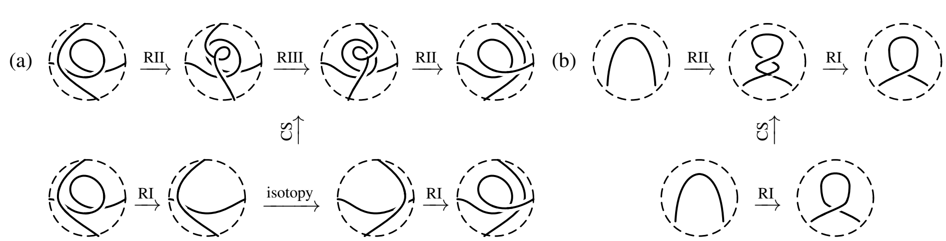


FIGURE 2. Two local Carter–Saito movie moves.

Definition 1.2.17 (Marked planar surfaces and tangles, [13, 2, 19])**.**

$$(F,a), \qquad F \subset \mathbb{R}^2 \text{ a smooth surface with boundary, closed in } \mathbb{R}^2.$$
$$a \subset \partial F \text{ an oriented 0-manifold}, \qquad |a| < \infty.$$
$$T \subset F \times \mathbb{R}, \qquad \dim T = 1, \qquad T \text{ compact and oriented.}$$
$$\partial T = T \cap (\partial F \times \mathbb{R}) = a \times \{0\} \quad \text{as oriented 0-manifolds.}$$

Definition 1.2.18 (Tangle isotopies and diffeomorphisms, [13, 2])**.**

$$T \simeq T' \iff \exists (T_s)_{s\in[0,1]} \subset F \times \mathbb{R}, \qquad \partial T_s = a \times \{0\}, \qquad (T_0, T_1) = (T, T'),$$
$$T \cong T' \iff \exists f\colon (F,a) \xrightarrow{\cong} (F',a'), \qquad (f \times \mathrm{Id}_{\mathbb{R}})(T) = T'.$$

Isotopies fix $a \times \{0\}$ pointwise; diffeomorphisms preserve orientations and carry the oriented points of a to those of a'.

Definition 1.2.19 (Tangle cobordisms, [14, 2, 19])**.**

$$S\colon T_0 \longrightarrow T_1, \qquad S \subset [0,1] \times F \times \mathbb{R}, \qquad \partial S = (-\{0\} \times T_0) \sqcup (\{1\} \times T_1) \sqcup ([0,1] \times a \times \{0\}),$$
$$S \cap ([0,1] \times \partial F \times \mathbb{R}) = [0,1] \times a \times \{0\}.$$

The surface is product-cylindrical on collars of every displayed boundary component.

$$\mathscr{E}_T := \{\text{interior planar isotopy}, \text{Reidemeister move}, \text{birth}, \text{death}, \text{planar saddle}\}.$$

Theorem 1.2.20 (Tangle complexes, [13, Theorem 2 and Proposition 27], [2, Theorems 1–3], [19, Lecture 2, Section 4(6)])**.**

$$(\partial F, a) \longmapsto A^t(\partial F, a), \qquad T \longmapsto \mathrm{CKh}^t(T) \in \mathrm{Kom}^b\big(\text{proj}-A^t(\partial F, a)\big).$$
$$A^t(S^1, \textit{two marked points}) = A^t, \qquad T \cong T' \overset{1.2.18}{\Longrightarrow} \mathrm{CKh}^t(T) \simeq \mathrm{CKh}^t(T').$$
$$(F,a) = (\mathbb{R}^2, \varnothing) \implies \mathrm{CKh}^t(T) = \textit{the complex of Definition 1.1.10}.$$

Proposition 1.2.21 (Tangle gluing, [13, Theorem 1 and Proposition 13])**.**

$$A^t(Z \sqcup Z', a \cup a') \cong A^t(Z,a) \otimes_{\mathbb{Z}[t]} A^t(Z',a').$$
$$F \cap F' = Z \subset \partial F \cap \partial F', \qquad a \cap Z = a' \cap Z \implies \mathrm{CKh}^t(T \cup_Z T') \simeq \mathrm{CKh}^t(T) \otimes_{A^t(Z, a\cap Z)} \mathrm{CKh}^t(T').$$

Proposition 1.2.22 (Elementary tangle maps, [13, Propositions 4–5, 7, and 28], [2, Theorems 1–3])**.**

$$S\colon T_0 \longrightarrow T_1, \qquad S \textit{ elementary} \overset{1.2.19}{\Longrightarrow} S_\#\colon \mathrm{CKh}^t(T_0) \longrightarrow \mathrm{CKh}^t(T_1).$$
$$S = \textit{an isotopy or Reidemeister trace} \implies S_\# \textit{ is a homotopy equivalence}.$$
$$(S \cup S')_\# \simeq S_\# \otimes S'_\#.$$

Proposition 1.2.23 (Mirror duality, [13, Proposition 30], [20, Propositions 5.7 and 5.9–5.10])**.**

$$T \subset D^2 \times \mathbb{R}, \qquad mT := \textit{reflection across } S^1 \implies \mathrm{CKh}^t(mT) \simeq q^{-|a|/2}\mathrm{Hom}_{A^t(S^1,a)}\big(\mathrm{CKh}^t(T), A^t(S^1,a)\big).$$

Theorem 1.2.24 (Strong functoriality up to sign, [11, Theorem 2], [14], [2, Theorems 4–5], [20])**.**

$$\Sigma, \Sigma'\colon L_0 \longrightarrow L_1, \qquad \Sigma \simeq_\partial \Sigma' \overset{1.2.16}{\iff} \mathscr{M}(\Sigma) \sim_{\mathrm{CS}} \mathscr{M}(\Sigma') \implies H(\Sigma_\#) = \pm H(\Sigma'_\#), \qquad \Sigma_\# \simeq \pm \Sigma'_\#.$$

Example 1.2.25 (A local movie-loop check, [20, Lemma 6.2])**.**

$$\mathscr{L} := \text{the left loop of Figure 2}, \qquad \mathscr{L}_\# \simeq \pm \mathrm{Id}.$$

Proof. Let T be the initial four-ended tangle of the left movie, set $A = A^t(S^1, \text{four points})$ and $C = \mathrm{CKh}^t(T)$; projectivity, duality, and gluing identify its endomorphism class.

$$[\mathscr{L}_\#] \in H_{0,0}\big(\mathrm{Hom}_A(C,C)\big),$$
$$\text{duality and gluing} \overset{1.2.20}{\Longrightarrow} H_{0,0}\big(\mathrm{Hom}_A(C,C)\big) \cong \mathrm{Kh}^t_{0,-2}(T\,mT),$$
$$T\,mT = U \sqcup U \implies H_{0,0}\big(\mathrm{Hom}_A(C,C)\big) \cong \mathrm{Kh}^t_{0,-2}(U \sqcup U) \cong \mathbb{Z}\cdot[\mathrm{Id}_C],$$
$$\mathscr{L} \text{ is a composition of Reidemeister equivalences} \overset{1.2.20}{\Longrightarrow} [\mathscr{L}_\#][\mathscr{L}_\#^{-1}] = [\mathrm{Id}_C],$$
$$H_{0,0}\big(\mathrm{Hom}_A(C,C)\big) = \mathbb{Z}\cdot[\mathrm{Id}_C] \implies [\mathscr{L}_\#] = \pm[\mathrm{Id}_C] \implies \mathscr{L}_\# \simeq \pm\mathrm{Id}_C.$$

Thus this local movie loop is chain homotopic to $\pm$Id. □

By Corollary 2.1.4, the sign ambiguity is removed in Subsection 2.1.

2. Exotic Surfaces, Signs, and Three-Spheres

This section follows Lee's orientation classes [17, 18], Rasmussen's cobordism maps [27], and the sign-corrected theory through exotic surfaces, sweep-around invariance, and dotted cobordisms. Throughout,

$$\mathbb{Q}_{t=1} := \mathbb{Q}[t]/(t-1), \qquad \mathscr{A}_\infty := [0,1]\times\{\infty\},$$

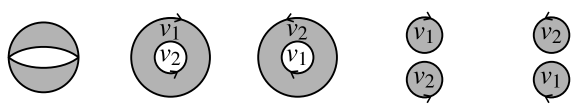
and undotted maps use the lasagna grading and Sano's coherent sign adjustment of the universal theory [29, Theorem 1 and Corollary 1.1].

2.1. **Orientation Generators and Corrected Maps.**

Definition 2.1.1 (Orientation generators, [17, 18, 27])**.** In $A^t \otimes_{\mathbb{Z}[t]} \mathbb{Q}_{t=1}$, set

$$v_1 := 1 + X, \qquad v_2 := 1 - X.$$

For $o \in \mathrm{Or}(L)$, checkerboard-color the oriented resolution with the unbounded region white, label a circle by v_1 exactly when its orientation agrees with the white-region boundary orientation, and by v_2 otherwise. Their tensor product is s_o.



Proposition 2.1.2 (The orientation generators are cycles)**.**

$$\partial s_o = 0 \qquad (o \in \mathrm{Or}(L)).$$

Proof. Every edge leaving the oriented resolution merges two distinctly labeled Seifert circles; write z_w for the tensor of unchanged labels.

$$v_o \longrightarrow w \overset{1.1.5}{\Longrightarrow} \mathscr{A}^t(v_o \to w)(s_o) = m(v_1 \otimes v_2) \otimes z_w,$$
$$m(v_1 \otimes v_2) = 0 \overset{1.1.1}{\Longleftrightarrow} (1+X)(1-X) = 1 - t = 0 \quad \text{in} \quad \mathbb{Q}_{t=1},$$
$$\mathscr{A}^t(v_o \to w)(s_o) = 0 \quad (v_o \to w) \overset{1.1.10}{\Longrightarrow} \partial s_o = \sum_{v_o \to w} \varepsilon(v_o, w)\mathscr{A}^t(v_o \to w)(s_o) = 0.$$

Thus s_o is a cycle. □

Theorem 2.1.3 (Rasmussen's orientation basis, [27, Theorem 2.2 and Proposition 4.1], [29, Proposition 3.4])**.**

$$H\big(\mathrm{CKh}^t(L)\otimes_{\mathbb{Z}[t]}\mathbb{Q}_{t=1}\big)\cong\bigoplus_{o\in\mathrm{Or}(L)}\mathbb{Q}\cdot[s_o].$$

For $\Sigma\colon L_0\to L_1$ such that every component meets L_0,

$$H\big(\Sigma_\#\otimes_{\mathbb{Z}[t]}\mathbb{Q}_{t=1}\big)([s_{o_0}])=\begin{cases}c_{\Sigma,o_0}[s_{o_1}], & o_0 \textit{ extends across } \Sigma,\\ 0, & o_0 \textit{ does not extend},\end{cases}\qquad c_{\Sigma,o_0}\in\{\pm 2^k\mid k\in\mathbb{Z}\}.$$

Corollary 2.1.4 (Blanchet–Sano sign correction, [3], [29, Theorem 1 and Corollary 1.1])**.** *The elementary signs admit a coherent choice satisfying*

$$\Sigma\simeq_\partial\Sigma'\overset{[29,\text{ Theorem 1}]}{\Longrightarrow}\Sigma_\#\simeq\Sigma'_\#,\qquad(\Sigma_2\circ\Sigma_1)_\#\simeq\Sigma_{2\#}\circ\Sigma_{1\#}.$$

Sano's distinguished orientation class fixes the remaining sign, and the correction survives every base change of the universal theory. Earlier coherent refinements appear in [5, 4].

2.2. **Closed Surfaces and Exotic Disks.**

Fact 2.2.1 (Closed-surface invariance, [26, 33], [29, Theorem 2])**.** *For connected closed oriented surfaces in* $[0,1]\times\mathbb{R}^3$*,*

$$g(\Sigma)=g(\Sigma')\implies\Sigma_\#=\Sigma'_\#.$$

Proposition 2.2.2 (Closed-surface evaluation, [26, 33], [29, Theorem 2])**.** *For a connected closed oriented* $\Sigma\subset[0,1]\times\mathbb{R}^3$ *with* $g(\Sigma)=g$*,*

$$\Sigma_\#(1)=\begin{cases}0, & g\text{ even},\\ 2^g\cdot t^{(g-1)/2}, & g\text{ odd}.\end{cases}$$

Proof. Replace Σ by the standard birth, g genus handles, and death.

$$m\Delta(1)=2\cdot X,\qquad m\Delta(X)=2\cdot t\overset{1.1.1}{\Longrightarrow}(m\circ\Delta)(a)=2\cdot Xa\quad(a\in A^t),$$
$$g(\Sigma)=g\overset{2.2.1}{\Longrightarrow}\Sigma_\#(1)=\varepsilon\big((m\circ\Delta)^g(1)\big)=2^g\cdot\varepsilon(X^g),$$
$$g\text{ even}\overset{1.1.1}{\Longrightarrow}X^g=t^{g/2},\qquad\varepsilon(X^g)=0,$$
$$g\text{ odd}\overset{1.1.1}{\Longrightarrow}X^g=t^{(g-1)/2}X\overset{1.1.1}{\Longrightarrow}\varepsilon(X^g)=t^{(g-1)/2}.$$
$$\varepsilon(X^g)\neq0\overset{1.1.1}{\Longleftrightarrow}g\text{ odd}.$$

Thus the displayed scalar is $\Sigma_\#(1)$. □

Example 2.2.3 (The 9_{46} disk movies, [31])**.** Figure 3 defines

$$K:=P(3,-3,3)=9_{46},\qquad\Sigma,\Sigma'\colon K\longrightarrow\varnothing.$$

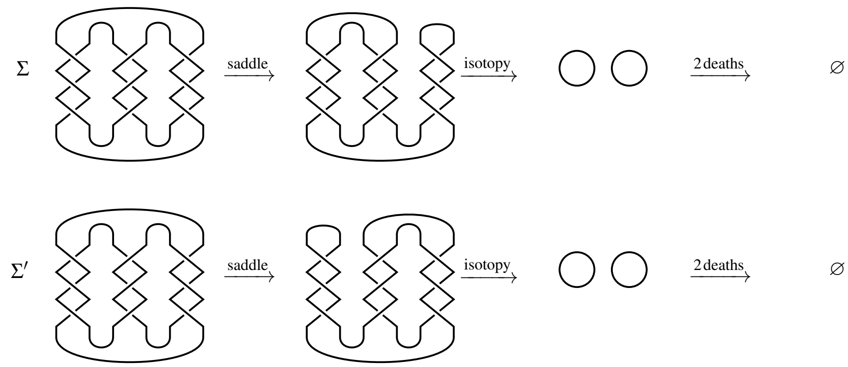


FIGURE 3. The knot 9_{46} and the two saddle-and-death movies defining the slice disks Σ and Σ'.

Figure 4 fixes the generators x, y, the saddle maps S, S', and the two-component unlink diagram D containing y.

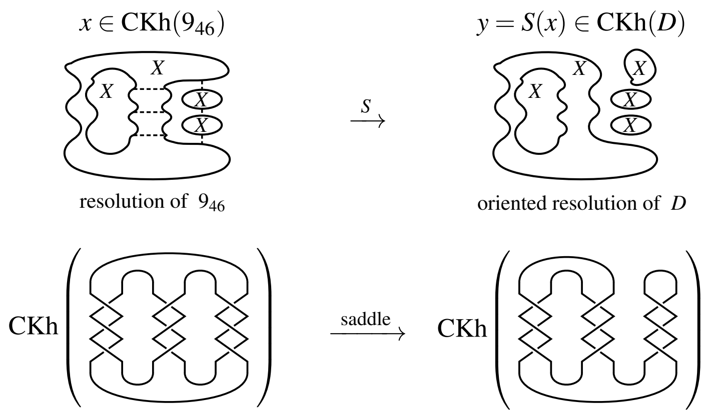


FIGURE 4. The generator $x \in \mathrm{CKh}(9_{46})$ and its saddle image $y = S(x) \in \mathrm{CKh}(D)$, with the six 1-resolution co-cores defining x drawn dashed.

Set

$$\mathrm{CKh} := \mathrm{CKh}^t \otimes_{\mathbb{Z}[t]} \mathbb{Z}[t]/(t).$$

Every edge leaving x merges two X-labeled circles, whence

$$t = 0 \overset{1.1.1}{\Longrightarrow} \Delta(X) = X \otimes X, \qquad m(X \otimes X) = 0,$$

$$m(X \otimes X) = 0 \overset{1.1.10}{\Longrightarrow} \partial x = 0, \qquad S(x) = y, \qquad S'(x) = 0,$$

$$(N_+, N_-) = (3, 6), \qquad |v_x| = |v_y| = 6 \overset{2}{\Longrightarrow} \mathrm{gr}_h(x) = \mathrm{gr}_h(y) = 6 - 6 = 0,$$

$$\#x^{-1}(X) - \#x^{-1}(1) = 4 \overset{1}{\Longrightarrow} \mathrm{gr}_q(x) = 3 - 12 + 6 + 4 = 1,$$

$$\#y^{-1}(X) - \#y^{-1}(1) = 5 \overset{1}{\Longrightarrow} \mathrm{gr}_q(y) = 3 - 12 + 6 + 5 = 2.$$

Proposition 2.2.4 (Non-isotopic slice disks for 9_{46}, [31])**.**

$$\Sigma \simeq \Sigma', \qquad H(\Sigma_\#)([x]) = \pm 1 \neq 0 = H(\Sigma'_\#)([x]) \overset{1.2.24}{\Longrightarrow} \Sigma \not\simeq_\partial \Sigma'.$$

Proof. Rotate the movies by π and evaluate their saddle-and-death maps on $[x]$.

$$R_\pi(\Sigma) = \Sigma' \implies \Sigma \simeq \Sigma',$$

$$[x] \in \mathrm{Kh}_{0,1}(K), \qquad H(S)([x]) = [y], \qquad H(S')([x]) = 0 \overset{2.2.3}{\implies} H(\Sigma'_\#)([x]) = 0.$$

For the orientation o after the saddle, y is the top quantum-filtered term of s_o.

$$[y] \in \mathrm{Kh}_{0,2}(D) \cong \mathbb{Z},$$

$$H(R_\#)([s_o]) = [s_{o'}], \qquad R_\# \colon \mathrm{CKh}(D) \simeq \mathrm{CKh}(U \sqcup U) \overset{[19,\text{ Section 2}]}{\implies} H(R_\#)([y]) = \pm[X \otimes X],$$

$$(\varepsilon \otimes \varepsilon)(X \otimes X) = 1 \overset{1.1.1}{\implies} H(\Sigma_\#)([x]) \in \{\pm 1\},$$

$$H(\Sigma_\#)([x]) = \pm 1 \neq 0 = H(\Sigma'_\#)([x]) \overset{1.2.24}{\implies} \Sigma \not\simeq_\partial \Sigma'.$$

Thus the two disks are isotopic only when the boundary is allowed to move. □

Fact 2.2.5 (Hayden–Sundberg exotic surfaces, [10, Theorem 1.1 and Lemma 4.4])**.** *For every $g \geq 0$, infinitely many knots bound genus-g surfaces satisfying*

$$\Sigma \simeq_{\mathrm{top},\partial} \Sigma', \qquad H(\Sigma_\#) \neq \pm H(\Sigma'_\#) \overset{1.2.24}{\implies} \Sigma \not\simeq_{\mathrm{sm},\partial} \Sigma'.$$

They may be chosen with

$$\mathrm{Sym}(K) = \{\mathrm{Id}\}, \qquad \Sigma \not\simeq_{\mathrm{sm}} \Sigma' \quad \textit{without fixing the boundary}.$$

Fact 2.2.6 (Exotic Seifert surfaces, [9, Theorems 1.2 and 3.3])**.** *Khovanov maps distinguish smoothly exotic Seifert surfaces in B^4.*

Fact 2.2.7 (One-stable exotic surfaces, [8, Theorem A], [7, Proposition E])**.** *Khovanov maps distinguish pairs remaining exotic after one internal stabilization.*

2.3. **Sweep-Around Invariance.**

Example 2.3.1 (The sweep-around move)**.** Figure 5 defines the sweep-around trace Sw_K by moving an arc adjacent to ∞ once around K.

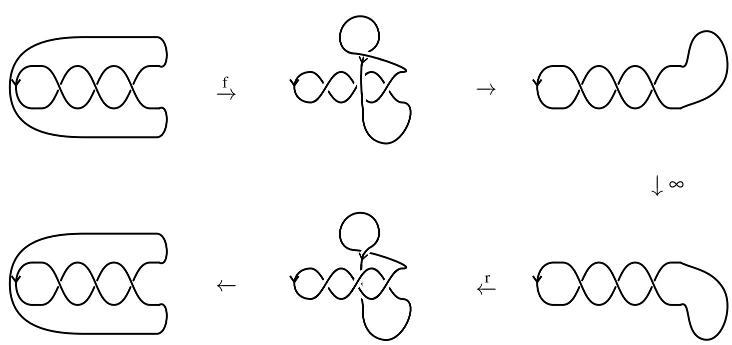


FIGURE 5. The oriented sweep-around isotopy of the positive trefoil, read across the top row, through ∞, and back across the bottom row.

$$\mathrm{Sw}_K \simeq_\partial [0,1] \times K \quad \text{in} \quad [0,1] \times S^3.$$

Open Problem 2.3.2 (Affine nontriviality)**.** The remaining claim is

$$\mathrm{Sw}_K \not\simeq_\partial [0,1] \times K \quad \text{in} \quad [0,1] \times \mathbb{R}^3.$$

Fact 2.3.3 (Sweep-around identity, [24, Theorem 1.1])**.** *For every link $L \subset \mathbb{R}^3$,*

$$H\big((\mathrm{Sw}_L)_\#\big) = \mathrm{Id}_{\mathrm{Kh}(L)}.$$

The account in [21, Proposition 3.7] *is corrected in* [25, Section 4.5].

Example 2.3.4 (A transverse passage through infinity)**.** For inversion in the unit sphere,

$$\iota(\mathbf{x}) := \frac{\mathbf{x}}{\|\mathbf{x}\|^2}, \qquad \iota(0) = \infty, \qquad \iota(\infty) = 0.$$

Figure 6 is the local model of a transverse passage through $\mathscr{A}_\infty$.

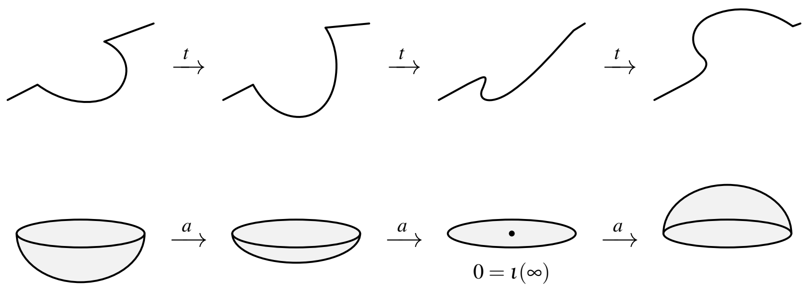


FIGURE 6. The family of swept surfaces crossing $[0,1] \times \{\infty\}$ once.

Theorem 2.3.5 (Sweep-around invariance, [24, Theorem 4.9])**.** *For link cobordisms $\Sigma_0, \Sigma_1 \subset [0,1] \times \mathbb{R}^3$,*

$$\Sigma_0 \simeq_\partial \Sigma_1 \quad \textit{in} \quad [0,1] \times S^3 \implies H((\Sigma_0)_\#) = H((\Sigma_1)_\#).$$

Proof. Perturb the parameterized isotopy Σ_a so its track is transverse to the parameterized axis at infinity, and choose $\varepsilon > 0$ isolating each intersection parameter.

$$\mathscr{S} := \bigcup_{a \in [0,1]} \{a\} \times \Sigma_a, \qquad \mathscr{A} := [0,1]_a \times [0,1]_t \times \{\infty\},$$

$$\mathscr{S} \pitchfork \mathscr{A} \implies \mathscr{S} \cap \mathscr{A} = \{p_1, \dots, p_r\},$$

$$0 < a_1 < \cdots < a_r < 1, \qquad a_i := \pi_a(p_i).$$

$$[b,c] \cap \{a_1, \dots, a_r\} = \varnothing \overset{1.2.16}{\implies} \Sigma_b \simeq_\partial \Sigma_c \quad \text{through movies in} \quad \mathbb{R}^3,$$

$$\Sigma_b \simeq_\partial \Sigma_c \overset{2.1.4}{\implies} H((\Sigma_b)_\#) = H((\Sigma_c)_\#),$$

$$p_i \overset{2.3.4}{\implies} \text{one sweep-around modification from } \Sigma_{a_i - \varepsilon} \text{ to } \Sigma_{a_i + \varepsilon} \overset{2.3.3}{\implies} H((\Sigma_{a_i-\varepsilon})_\#) = H((\Sigma_{a_i+\varepsilon})_\#),$$

$$H((\Sigma_0)_\#) = H((\Sigma_{a_1-\varepsilon})_\#) = H((\Sigma_{a_1+\varepsilon})_\#) = \cdots = H((\Sigma_1)_\#).$$

Thus every transverse passage through infinity preserves the map. □

Lemma 2.3.6 (Normal correction along the axis, [24, proof of Lemma 4.7])**.** *Let U_∂ be a boundary collar and*

$$\Phi \in \mathrm{Diff}^+([0,1] \times S^3), \qquad \Phi|_{U_\partial \cup \mathscr{A}_\infty} = \mathrm{Id}.$$

There are neighborhoods $W \subset V$ of $\mathscr{A}_\infty$ and $\Psi, \widehat{\Phi} \in \mathrm{Diff}^+([0,1] \times S^3)$ satisfying

$$\mathrm{supp}(\Psi) \subset V, \qquad \Psi|_{U_\partial \cup \mathscr{A}_\infty} = \mathrm{Id}, \qquad d_\nu \Psi = d_\nu \Phi,$$

$$\Phi \circ \Psi^{-1} \simeq_\partial \widehat{\Phi}, \qquad \widehat{\Phi}|_W = \mathrm{Id}.$$

For every fixed surface F,

$$F \cap \mathscr{A}_\infty = \varnothing \implies \mathrm{supp}(\Psi) \cap F = \varnothing.$$

Proof. Trivialize $\nu\mathscr{A}_\infty$ and polar-decompose the based normal derivative loop.

$$B_t := d_\nu\Phi_t = Q_tP_t, \qquad B_0 = B_1 = \mathrm{Id}, \qquad Q_t \in \mathrm{SO}(3), \qquad P_t > 0.$$

$$\{P \in \mathrm{GL}^+(3,\mathbb{R}) \mid P = P^\mathsf{T} > 0\} \simeq \{\mathrm{Id}\} \overset{\text{[24, proof of Lemma 4.7]}}{\Longrightarrow} \exists\Psi_P \quad \mathrm{supp}(\Psi_P) \subset V, \quad \Psi_P|_{U_\partial \cup \mathscr{A}_\infty} = \mathrm{Id}, \quad d_\nu\Psi_P = P_t,$$
$$[Q_t] \in \pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}/2\mathbb{Z},$$
$$[Q_t] = 0 \overset{\text{[24, proof of Lemma 4.7]}}{\Longleftrightarrow} \exists Q_{t,s} \quad Q_{t,0} = Q_t, \quad Q_{t,1} = \mathrm{Id}, \quad Q_{0,s} = Q_{1,s} = \mathrm{Id},$$
$$\chi\colon [0,1] \longrightarrow [0,1], \qquad \chi(r) = 0 \text{ near } r = 0, \qquad \chi(r) = 1 \text{ near } r = 1,$$
$$\Psi_Q(t, r\mathbf{v}) = \big(t, rQ_{t,\chi(r)}\mathbf{v}\big), \qquad d_\nu\Psi_Q = Q_t.$$
$$[Q_t] = 1 \overset{\text{[24, proof of Lemma 4.7]}}{\Longrightarrow} \exists T \quad [d_\nu T] = 1, \quad T|_{\mathscr{A}_\infty} = \mathrm{Id}, \quad \mathrm{supp}(T) \subset B^4,$$
$$[Q_t(d_\nu T)^{-1}] = 0, \overset{\text{[24, proof of Lemma 4.7]}}{\Longrightarrow} \exists\Psi_0, \qquad d_\nu\Psi_0 = Q_t(d_\nu T)^{-1}, \qquad \Psi_Q := \Psi_0 \circ T.$$
$$\Psi := \Psi_Q \circ \Psi_P, \qquad d_\nu\Psi = Q_tP_t = B_t, \qquad d_\nu(\Phi \circ \Psi^{-1}) = \mathrm{Id},$$
$$d_\nu(\Phi \circ \Psi^{-1}) = \mathrm{Id} \overset{\text{[24, proof of Lemma 4.7]}}{\Longrightarrow} \Phi \circ \Psi^{-1} \simeq_\partial \widehat{\Phi}, \qquad \widehat{\Phi}|_W = \mathrm{Id},$$
$$F \cap \mathscr{A}_\infty = \varnothing \implies V \cap F = \varnothing \implies \mathrm{supp}(\Psi) \cap F = \varnothing.$$

Thus Ψ corrects the normal derivative without meeting F. ☐

Theorem 2.3.7 (Diffeomorphism invariance, [24, Lemma 4.7])**.**

$$\Phi \in \mathrm{Diff}^+([0,1] \times S^3), \qquad \Phi|_{\partial([0,1]\times S^3)} = \mathrm{Id}, \qquad \Sigma\colon L_0 \longrightarrow L_1 \implies H(\Sigma_\#) = H((\Phi(\Sigma))_\#).$$

Proof. Perturb Σ off the axis, straighten the boundary collar and image axis, correct the normal derivative, and dilate Σ into the fixed region.

$$\Phi \simeq_\partial \Phi_0, \qquad \Phi_0|_U = \mathrm{Id}, \qquad \Phi_0(\mathscr{A}_\infty) = \mathscr{A}_\infty, \qquad \Phi_0|_{\mathscr{A}_\infty} = \mathrm{Id},$$
$$\Sigma \cap \mathscr{A}_\infty = \varnothing \overset{2.3.6}{\Longrightarrow} \mathrm{supp}(\Psi) \cap \Sigma = \varnothing, \qquad \Phi_0 \circ \Psi^{-1} \simeq_\partial \widehat{\Phi}, \qquad \widehat{\Phi}|_W = \mathrm{Id},$$
$$\Psi^{-1}(\Sigma) = \Sigma, \qquad \widehat{\Phi}(\Sigma) \simeq_\partial \Phi_0(\Sigma), \qquad \Phi_0(\Sigma) \simeq_\partial \Phi(\Sigma),$$
$$\ell := [0,1] \times \{p\}, \qquad \ell \cap \Sigma = \varnothing, \qquad \Sigma_0 = \Sigma, \qquad \Sigma_1 \subset U \cup W, \qquad \widehat{\Phi}(\Sigma_1) = \Sigma_1,$$
$$\Sigma = \Sigma_0 \simeq_\partial \Sigma_1 = \widehat{\Phi}(\Sigma_1) \simeq_\partial \widehat{\Phi}(\Sigma_0) \simeq_\partial \Phi_0(\Sigma) \simeq_\partial \Phi(\Sigma),$$
$$\Sigma \simeq_\partial \Phi(\Sigma) \quad \text{in} \quad [0,1] \times S^3 \overset{2.3.5}{\Longrightarrow} H(\Sigma_\#) = H((\Phi(\Sigma))_\#).$$

Thus Φ preserves the Khovanov map. ☐

2.4. **Dotted Cobordisms.**

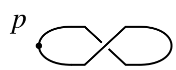
Definition 2.4.1 (Dotted cobordisms, [2, Section 11.2])**.** A *dotted cobordism* is

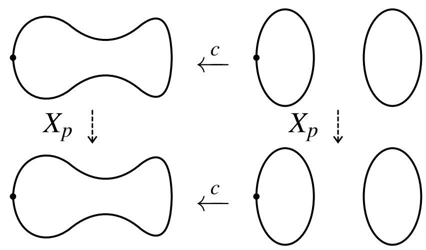
$$(\Sigma, P), \qquad \Sigma\colon L_0 \longrightarrow L_1, \qquad P \in \mathrm{Sym}^k(\mathrm{int}(\Sigma)).$$

A dot p on a resolution circle acts by multiplication by X on its A^t-factor, and

$$\Sigma_{\#,P}\colon \mathrm{CKh}^t_{i,j}(L_0) \longrightarrow \mathrm{CKh}^t_{i,j-\chi(\Sigma)+2k}(L_1).$$

Example 2.4.2 (The one-crossing dotted unknot)**.** The dotted identity X_p of the one-crossing unknot acts on both resolutions as follows.





$$\begin{gathered} X_p(1) = X, \qquad X_p(X) = t, \\ X_p(1 \otimes 1) = X \otimes 1, \qquad X_p(1 \otimes X) = X \otimes X, \\ X_p(X \otimes 1) = t \cdot (1 \otimes 1), \qquad X_p(X \otimes X) = t \cdot (1 \otimes X). \end{gathered}$$

Proposition 2.4.3 (Strong functoriality for dotted cobordisms, [2, Section 11.2])**.**

$$(\Sigma, P) \simeq_\partial (\Sigma', P') \implies \Sigma_{\#,P} \simeq \pm \Sigma'_{\#,P'}.$$

Proof. Let $\mathscr{M}, \mathscr{M}'$ be generic dotted movies; their isotopy adds dot slides and distant commutations to the Carter–Saito moves.

$$\begin{gathered} (\Sigma, P) \simeq_\partial (\Sigma', P') \overset{1.2.16}{\implies} \mathscr{M} \sim_{\mathrm{CS,\,dot}} \mathscr{M}', \\ \mathrm{supp}(X_p) \cap \mathrm{supp}(F) = \varnothing \implies X_p F = F X_p, \qquad X_p X_q = X_q X_p, \\ m(Xa \otimes b) = X\, m(a \otimes b) = m(a \otimes Xb), \\ \Delta(Xa) = (X \otimes 1)\Delta(a) = (1 \otimes X)\Delta(a) \overset{2.4.1}{\implies} X_p \circ S = S \circ X_p, \\ p, p' \text{ separated by one crossing } c \overset{\text{[2, Section 11.2]}}{\implies} X_{p'} \simeq \pm X_p. \\ \mathscr{M} \sim_{\mathrm{CS,\,dot}} \mathscr{M}' \overset{\text{1.2.24, [2, Section 11.2]}}{\implies} \mathscr{M}_\# \simeq \pm \mathscr{M}'_\# \implies \Sigma_{\#,P} \simeq \pm \Sigma'_{\#,P'}. \end{gathered}$$

Thus dotted isotopy changes the chain map by at most one overall sign. $\square$

Part 2. **Exercises**

3. Exercises

Exercise 3.1 (1.1.3)**.** Show

$$(A^t, m, \Delta, 1, \varepsilon) \text{ is TQFT-graded Frobenius over } \mathbb{Z}[t], \qquad \mathrm{gr}_q(1) = -1, \quad \mathrm{gr}_q(X) = 1, \quad \mathrm{gr}_q(t) = 4.$$

Exercise 3.2 (1.1.7)**.** Show

$$\text{every two-dimensional face of } \mathscr{A}^t \text{ commutes.}$$

Exercise 3.3 (1.1.9)**.** Show

$$\text{every two-dimensional face of } \mathscr{A}^t_\varepsilon \text{ anticommutes.}$$

Exercise 3.4 (1.1.11)**.** For two crossing orderings, show

$$\mathrm{CKh}^t_\varepsilon(L) \cong \mathrm{CKh}^t_{\varepsilon'}(L) \quad \text{bigraded.}$$

Exercise 3.5 (1.1.14)**.** Show

$$D' = \mathrm{RI}(D) \implies \mathrm{CKh}^t(D') \simeq \mathrm{CKh}^t(D) \quad \text{bigraded.}$$

Exercise 3.6 (1.1.13)**.** Show

$$(|v|, N_+, N_-) \longmapsto (|v|+1, N_+ + 1, N_- + 1) \implies (\mathrm{gr}_h, \mathrm{gr}_q) \longmapsto (\mathrm{gr}_h, \mathrm{gr}_q).$$

Exercise 3.7 (1.2.3)**.** Show there exist embedded cobordisms

$$\Sigma_{\mathrm{b}}, \qquad \Sigma_{\mathrm{d}}, \qquad \Sigma_{\mathrm{s}}$$

representing a birth, death, and planar saddle.

Exercise 3.8 (1.2.4)**.** Show

$$\Sigma \colon L_0 \longrightarrow L_1 \implies \Sigma \simeq_\partial \Sigma_n \circ \cdots \circ \Sigma_1, \qquad \Sigma_i \in \mathscr{E}.$$

Exercise 3.9 (1.2.6)**.** Show

$$U \cap D = \varnothing, \qquad N(U) = 0 \implies \mathrm{CKh}^t(D \sqcup U) \cong \mathrm{CKh}^t(D) \otimes_{\mathbb{Z}[t]} A^t.$$

Exercise 3.10 (1.2.10)**.** Show

$$\mathrm{Kh}^t(K; \mathbb{Q}[t]) \cong \mathbb{Q}[t]_{(0,p)} \oplus \mathbb{Q}[t]_{(0,q)} \oplus \mathrm{tors}_{\mathbb{Q}[t]}\big(\mathrm{Kh}^t(K; \mathbb{Q}[t])\big).$$

Exercise 3.11 (1.2.13)**.** Show

$$\Sigma \colon K_1 \longrightarrow K_2, \qquad |\pi_0(\Sigma)| = 1, \qquad g(\Sigma) = g \implies |s(K_1) - s(K_2)| \leq 2g.$$

Exercise 3.12 (1.2.14)**.** Show

$$\gcd(p,q) = 1, \qquad p, q > 0 \implies s(T_{p,q}) = (p-1)(q-1), \qquad g_4(T_{p,q}) = \frac{(p-1)(q-1)}{2}.$$

Exercise 3.13 (1.2.19)**.** Define

$$T \simeq T', \qquad T \cong T', \qquad S \colon T_0 \longrightarrow T_1, \qquad S \text{ elementary.}$$

Exercise 3.14 (2.1.2)**.** Show

$$\partial s_o = 0.$$

Exercise 3.15 (2.2.2)**.** For a connected closed oriented surface Σ of genus g, show

$$\Sigma_\#(1) = \begin{cases} 0, & g \text{ even,} \\ 2^g t^{(g-1)/2}, & g \text{ odd.} \end{cases}$$

Exercise 3.16 (2.3.6)**.** Set

$$\mathscr{A} := [0,1] \times \{\infty\}.$$

Show there exist a correction Ψ supported near $\mathscr{A}$ and a neighborhood $V \supset \mathscr{A}$ such that

$$d_V\Psi = d_V\Phi, \qquad (\Phi \circ \Psi^{-1})|_V = \mathrm{Id}.$$

Exercise 3.17 (2.4.3)**.** Show

$$(\Sigma, P) \simeq_\partial (\Sigma', P') \implies \Sigma_{\#,P} \simeq \pm\Sigma'_{\#,P'}.$$

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